

Operating Systems

2520030227

N.Prudvi Raj

SEC-5

WEEK-3

3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

Code

```
prudvi@LAPTOP-TEKO376B: ~ × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
    }
    else if (pid == 0)
    {
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());

        sleep(10);

        printf("Child finished\n");
    }
    else
    {
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);

        sleep(5);

        wait(NULL);

        printf("Parent finished\n");
    }

    return 0;
}
```

output

```
prudvi@LAPTOP-TEKO376B:~$ nano fork.c
prudvi@LAPTOP-TEKO376B:~$ gcc fork.c
prudvi@LAPTOP-TEKO376B:~$ ./a.out
Parent Process
PID = 603
Child PID = 604
Child Process
PID = 604
PPID = 603
Child finished
Parent finished
```